

## **Lab 2: UART transmission with the ATmega328P/PB**

### **Introduction**

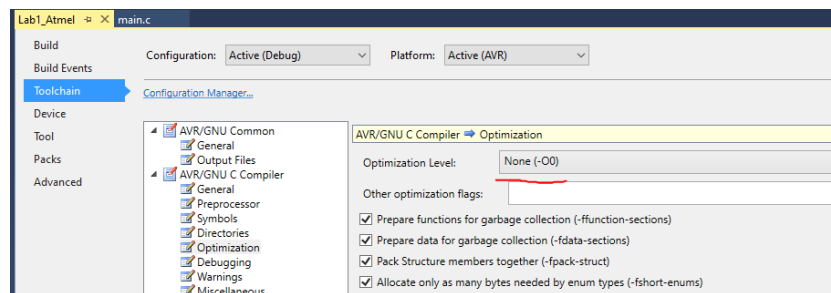
The objective of this lab is to introduce you to data transmission, using the UART protocol and polling. You will also develop experience using Atmel/Microchip Studio and the Proteus Virtual System Modelling (VSM) tool to develop and debug your embedded software. You will test the functionality of your code using Proteus VSM, which emulates a virtual ATmega328P microcontroller and associated peripherals. Subsequently you will test the code you develop on an ATmega328P /ATmega328PB Xplained Mini evaluation board.

**Read the entire lab manual before you start.** This lab should take you approximately **4 hours**. There is a pre-lab, five compulsory parts and a few optional activities. **You should aim to complete at least the pre-lab before attending the support session. The final answers should be written in this document, with supporting calculations shown in your logbook.** Before starting the lab, pull the lab repository from GitHub as it contains the Proteus model(s). Remember to commit and push your saved work (code and Proteus simulations) to the lab repository on GitHub after completing each task.

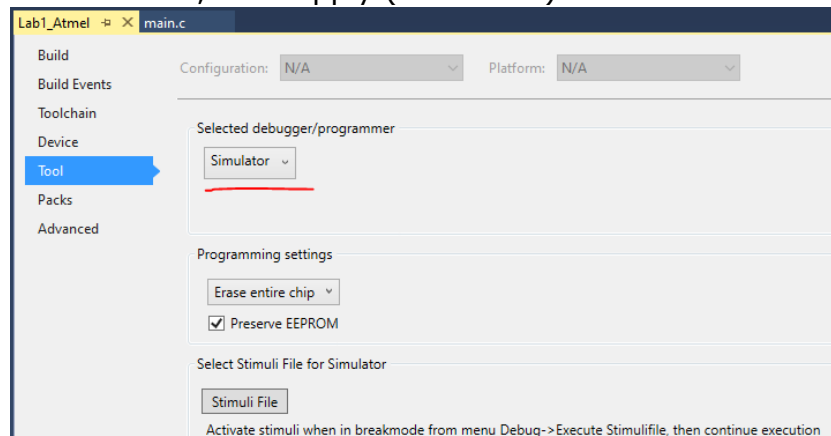
### **Pre-Lab: Using Atmel Studio**

In this course we are using the ATmega328P in Proteus VSM (since it only supports this MCU) and the ATmega328P/ATmega328PB when it comes to testing your code on actual hardware. ATmega328P behaves almost the same as the ATmega328PB you already encountered in COMPSYS201. The code written for an ATmega328P could be directly compiled to work on the ATmega328PB (all most all the time). Before the lab, we want to make sure that you are familiar with how to use Atmel/Microchip Studio to write some code, compile, and simulate. Referring to your CS201 coursebook where necessary, complete these tasks:

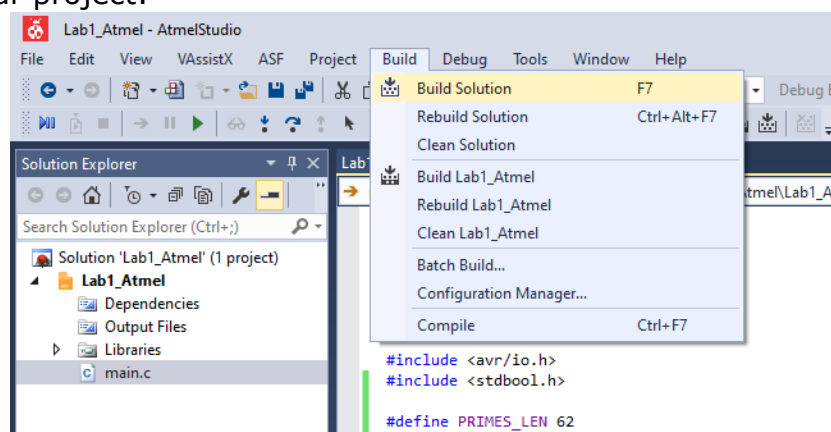
- 1) Create a new GCC C Executable Project in Atmel/Microchip Studio, selecting ATmega328P as the device family (note the new version of Atmel Studio is called Microchip Studio).
- 2) Write a piece of code that creates an array of prime numbers up to 300. HINT: Refer to lecture notes - there should be 62 prime numbers in total. You can use a simple algorithm that iterates through all the numbers up to 300, and checks if they can be exactly divided by numbers smaller than it using the modulo operator. This code should only be executed once, so write your code *before* the while(1) loop. Make sure to use an appropriate variable type for storing all of the numbers in this array.
- 3) Before building the program, go to Project > [ProjectName] Properties > Toolchain > AVR/GNU C Compiler > Optimization, and set the Optimization level to "None -O0". This ensures that the compiler doesn't remove any of your code that it thinks isn't being used:



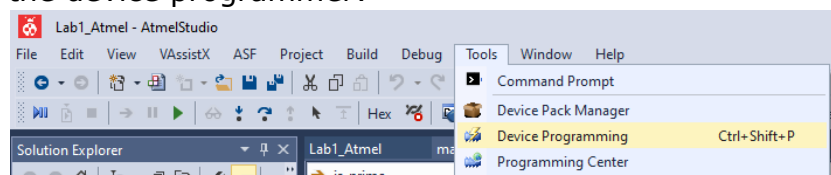
- 4) Build the program, and then go to Tools > Device Programming, select Simulator for the Tool, click Apply (see below):



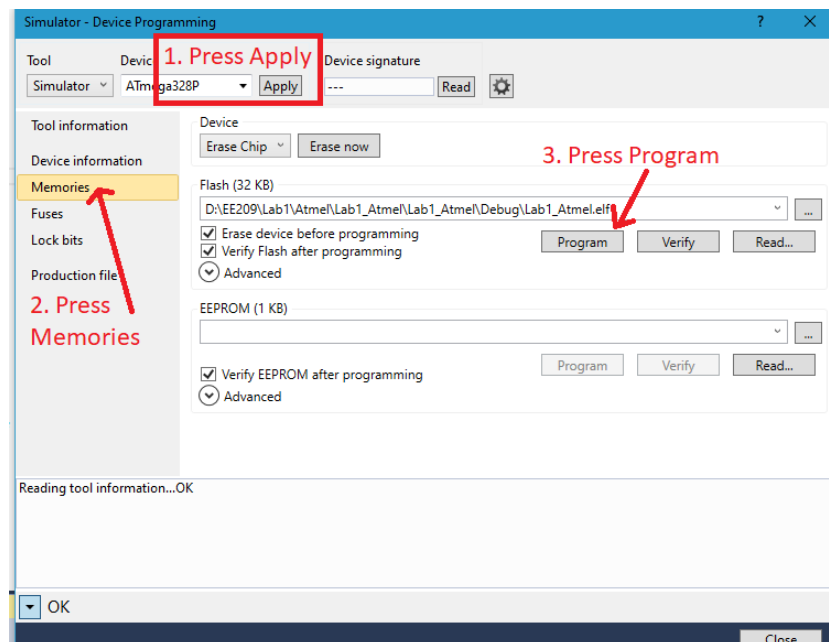
- 5) Build your project:



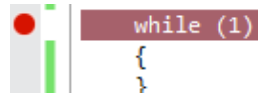
- 6) Bring up the device programmer:



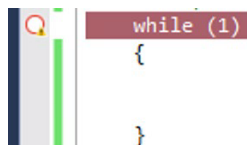
- 7) Press "Apply" with "Simulator" active, press "Memories", and click "Program":



- 8) Press "Close" after you've pressed Program.
- 9) Set up a breakpoint next to the "while(1)" loop. You can either click on the left-hand margin next to the code on the appropriate line or right click on the "while(1)" and select Breakpoint>Insert Breakpoint. A dark red circle should appear to indicate that there is a breakpoint here:

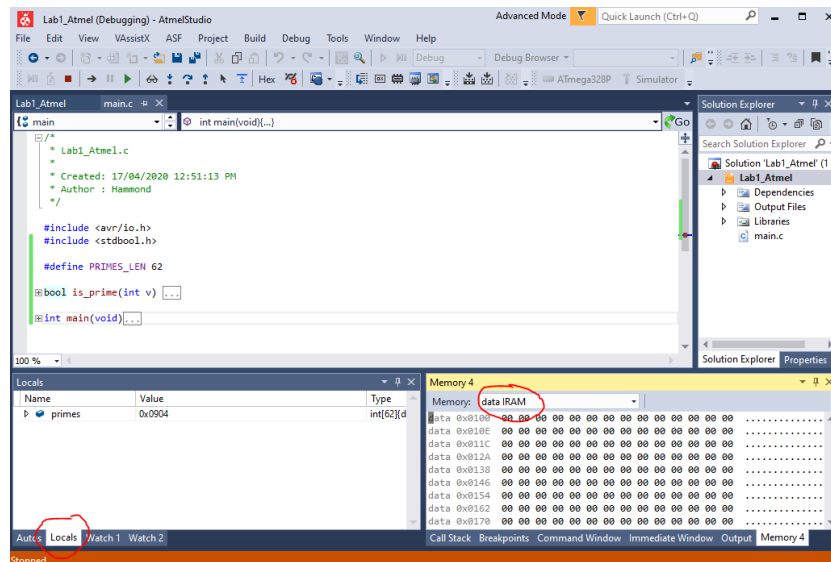


**Make sure there are NO blank lines between the two brackets of while(1) as shown above.** If you had by mistake inserted blank lines between the two bracket of while(1), when you start debugging as per the next step below, the dark red circle will become a white circle with a red boarder and a warning will be shown as in:



If this is the case, delete the blank lines, build and program as per previous steps.

- 10) Press the  button to "Start Debugging and Break" or press (Alt + F5). This should execute the code and open a couple of new windows. In the bottom left, you should go to the "Locals" tab to see your local variables, such as your array of prime numbers (our array is named *primes* in this example). On the bottom right, you should change the Memory window from "prog FLASH" to "data IRAM". Here, you can see the raw hexadecimal values of what would be in the memory of the microcontroller:



- 11) Press "Continue" (it may take a few minutes to simulate - so be patient):



- 12) Using the memory location given in the "Locals" window (under "Value", e.g. for us 0x0880), find the corresponding memory address in the Memory window. Verify that both of the "Locals" and "Memory" windows show you a list of 62 prime numbers up to 300. Confirm that both windows are actually showing you the same numbers in decimal and hex (we include an example here – note that your memory addresses may differ):

| Locals |        |                                     |
|--------|--------|-------------------------------------|
| Name   | Value  | Type                                |
| primes | 0x0880 | uint16_t[62](data)@0x0880 ([R28]+7) |
| [0]    | 2      | uint16_t(data)@0x0880               |
| [1]    | 3      | uint16_t(data)@0x0882               |
| [2]    | 5      | uint16_t(data)@0x0884               |
| [3]    | 7      | uint16_t(data)@0x0886               |
| [4]    | 11     | uint16_t(data)@0x0888               |

| Memory 4    |                                                                            |                      |
|-------------|----------------------------------------------------------------------------|----------------------|
| Memory:     | data IRAM                                                                  | Address: 0x0880,data |
| data 0x0880 | 02 00 03 00 05 00 07 00 0b 00 0d 00 11 00 13 00 17 00 1d 00 1f 00 25 00 29 |                      |
| data 0x0899 | 00 2b 00 2f 00 35 00 3b 00 3d 00 43 00 47 00 49 00 4f 00 53 00 59 00 61 00 |                      |

- 13) While the debugger is running, at the top of the code window you should also see a new Disassembly tab. You may be interested in having a look at this, which shows the assembly code that your C code has compiled to. When you are finished, click "Stop Debugging" to return to your code.

$$256 + 2 \times 16 \times 5$$

Q P.1: What was the largest number stored in the array, in decimal and in hex?

293      0x0125

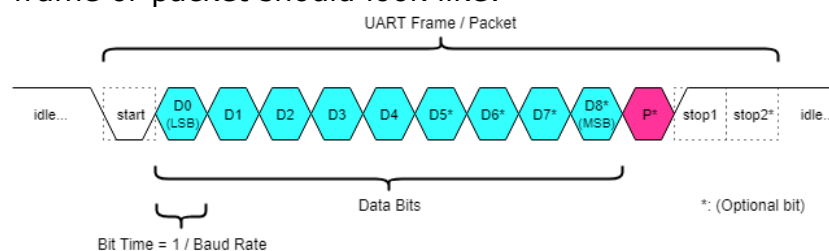
Q P.2: Why did the memory window show a lot of 00 elements between some of the numbers in the array?

Each element of the array is a 16 bit uint16\_t value, so it occupies 2 bytes. Since most prime numbers are smaller than 256, their upper byte is 0x00.

## Part 1: What is UART?

Let us imagine that we have two computers, and they need to talk to each other. We want to get data from Computer A to Computer B, so we form a connection between them. Great! But what do we send down that connection?

Assume that computer A is trying to talk to computer B. Computer A could send a bunch of 0 and 1 bits over a period of time in a *serial* manner (i.e. one bit at a time). If Computer B doesn't know how long each bit is going to be or how much data there is or when to start reading, then it cannot understand what Computer A is sending to it. So, we need a *communication protocol*, a set of rules that Computer A and Computer B both agree upon in order to know how to transmit and receive the data. UART is one such protocol (among others discussed in lectures), and it has a number of *parameters* which allow the same protocol to be used in slightly different ways for different purposes. This diagram shows what a single UART frame or packet should look like:



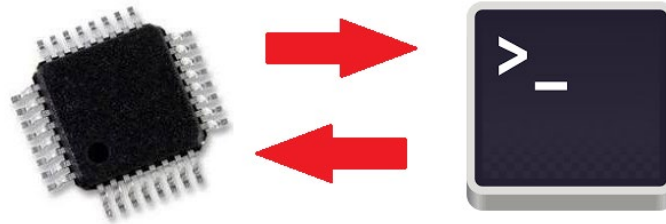
Q 1.1: What is the purpose of each of the following parameters and bits?

- i) The start bit indicates the beginning of a new UART frame and allows the receiver to synchronise with the incoming data.
- ii) The parity bit The parity bit provides simple error detection by indicating whether the number of 1 bits should be odd or even.
- iii) The stop bits it indicates the end of a UART frame and returns the communication line to its idle state.
- iv) The baud rate it specifies the number of transmitted symbols or bits per second and therefore determines the duration of each UART bit.

In order for two devices to communicate over UART, all parameters used by both devices need to match. For this lab (and in the project), we will use these values:

|             |      |             |      |
|-------------|------|-------------|------|
| Baud Rate   | 9600 | Parity Bit  | None |
| # Data Bits | 8    | # Stop Bits | 1    |

The aim of this lab is to send some data out of your Microcontroller via the UART. In the real world, any device could be listening, whether that be a computer, a cell phone, another microcontroller, etc. In this initial part of this lab, we'll be sending data to Proteus' *virtual terminal*. After you verify the functionality of your program in Proteus, you load this program to the Xplained Mini that's connected to a computer through the USB cable. The data coming out of the microcontroller is then viewed on the computer using a terminal program such as PuTTY or RealTerm:



On the ATmega328P microcontroller, there is a USART module (the ATmega328PB has an additional USART module, which is one of the very few differences). This is a standalone module whose sole purpose is to transmit and receive data according to the USART protocols. This module knows all of the rules already, so we only need to tell the module two things: what parameter values we want it to use (**control**) and what we want to transmit (**data**).

Note: UART stands for Universal Asynchronous Receiver Transmitter – the extra S in USART stands for Synchronous, because the USART module can be used to also follow a synchronous protocol. However, for the purposes of this project, we only have a single communication line, so we cannot send a synchronising clock along with our data to the receiver side. There are better synchronous communication protocols anyway, like SPI and I<sup>2</sup>C, but UART is the most popular asynchronous protocol. So wherever you see USART in the datasheet or online, in most cases it is the same thing as UART.

## Part 2: How to Control the Microcontroller?

We primarily interact with the I/O peripherals (also known as hardware modules) on a microcontroller through its *registers*. Some registers just store data, while the vast majority are used to control how the peripheral behaves. In order to control a peripheral, there are a bunch of settings that need to be configured. Most settings only need one or two bits to configure, so a single register often manages more than one setting. The ATmega328P registers are 8-bit (hence it is referred to as an “8-bit” microcontroller). There are five 8-bit control registers and one 8-bit data register associated with the USART peripheral on this microcontroller.

*Q 2.1: Using the datasheet, state the register and bits that control each of the settings below, and describe the purpose of each of these settings. Identify whether or not we need this setting for this lab, and if we do, determine if the setting is set at initialisation or if it is used at runtime:*

| Setting Name           | Register and Bits          | Purpose                                                                                      | Do we need it? | Initialisation/ Runtime |
|------------------------|----------------------------|----------------------------------------------------------------------------------------------|----------------|-------------------------|
| Receive Complete       | UCSR0A<br>RXC0<br>(Bit 7)  | A flag indicating when the USART has received a complete packet that is ready to be read     | No             | -                       |
| Tx Data Register Empty | UCSR0A<br>UDRE0<br>(Bit 5) | A flag indicating that the USART is ready to send another packet and it is safe to load data | Yes            | Runtime                 |

|                    |                                        |                                                                              |     |                 |
|--------------------|----------------------------------------|------------------------------------------------------------------------------|-----|-----------------|
| Transmit Complete  | UCSR0A<br>TXC0<br>(bit 6).             | Set a transmit complete flag after transmitting a data frame.                | NO  | -               |
| Mode Selection     | UCSR0C<br>UMSEL0[1..0]<br>(bit 6 & 7). | Mode select where<br>00 = UART. 01 = USART.<br>10 = Reserved. 11 = SPI.      | YES | Initialisation. |
| Character Size     | UCSR0C<br>UCSZ0[1..0]<br>(bit 1 & 2).  | Character Size where<br>00 = 5-bits. 01 = 6-bits<br>10 = 7-bits. 11 = 8-bits | YES | Initialization  |
| Clock Polarity     | UCSR0C<br>UCPOL0<br>(bit 0).           | Sets clock polarity for synchronous USART mode                               | No  | -               |
| Baud Rate          | UBRR0H &<br>UBRR0L                     | Sets the USART baud-rate divider                                             | YES | Initialisation  |
| Receiver Enable    | UCSR0B<br>RXEN0<br>(bit 4)             | Enables the USART receiver                                                   | No  | -               |
| Transmitter Enable | UCSR0B<br>TXEN0<br>(bit 3)             | Enables the USART transmitter                                                | YES | Initialisation  |
| Parity Mode        | UCSR0C<br>UPM0[1..0]<br>(bit 5 & 4)    | Enables and selects parity type                                              | YES | Initialisation  |
| Parity Error       | UCSR0A<br>UPF0<br>(bit 2)              | Indicates a parity error in received data                                    | NO  | -               |

$$UBRR = \frac{f_{osc}}{16 \times BAUD} - 1$$

One of the most important USART parameters is the *baud rate*. Inside the USART peripheral, there is essentially a clock divider that is able to convert the system clock into a clock that matches the desired baud rate. On the ATmega328P, this value is stored in the UBRR register (USART Baud Rate Register).

Q 2.2: Using the datasheet, what is the appropriate UBRR value? Note that our system clock is 2MHz. UBRR = 12.

Q 2.3: On the ATmega328P, the UBRR is divided up between two registers, UBRR0H and UBRR0L. Why do you think this is the case?

The UBRR value is 12 bits wide, while the ATmega328P uses the 8-bit register. It must be split between a high & low register.

Other than the two UBRR registers, there are three more control registers. We need to decide what values to set them to so that we get the behaviour that we expect/need from the USART peripheral. This is referred to as initialising a peripheral and is typically done at the beginning of your embedded program.

Q 2.4: Using your answers from Q 2.1 and the datasheet, for each of the three control registers below, fill out the table with the short name for each bit, and the value we need to set it to during initialisation. Your options are 0 or 1 – if you don't care what the value is, set it to 0.

**UCSR0A**

|      |      |       |     |      |      |      |       |
|------|------|-------|-----|------|------|------|-------|
| RXC0 | TXC0 | UDR20 | F20 | DOR0 | UP20 | U2X0 | MPCM0 |
| 0    | 0    | 0     | 0   | 0    | 0    | 0    | 0     |

**UCSR0B**

|        |        |        |       |       |        |       |       |
|--------|--------|--------|-------|-------|--------|-------|-------|
| RXCIF0 | TXCIF0 | UDRIF0 | RXEN0 | TXEN0 | UCSZ02 | RXIF0 | TXIF0 |
| 0      | 0      | 0      | 0     | 1     | 0      | 0     | 0     |

**UCSR0C**

|         |         |       |       |        |        |        |       |
|---------|---------|-------|-------|--------|--------|--------|-------|
| UMSZL01 | UMSZL00 | UPM01 | UPM00 | USIS00 | UCSZ01 | UCSZ00 | ULP00 |
| 0       | 0       | 0     | 0     | 0      | 1      | 1      | 0     |

During initialisation, we can directly write these binary values to the three USART control registers. The following example shows how we could set UCSR0A to have 00100101, assuming we have <avr/io.h> included in the project.

```
UCSR0A = 0b00100101; //This is an arbitrary value
```

The 0b means that the number is in binary. Sometimes during system operation, we may need to read or write to a specific bit. Thinking back to COMPSYS201, there were three types of instructions that we needed to implement – reading, writing 1s, and writing 0s. Thanks to the AVR SDK (Software Development Kit), we have access to a bunch of macros in avr/io.h, which can make our life slightly easier.

Instead of using integers that don't really mean anything to a human reader, we can use the actual names of the bits instead. For example, we can use (1<<RXC0) instead of (1<<7); as the human-readable name is more *meaningful*. Here RXC0 refers to the position of the RXC0 bit in the UCSR0A control register, which is 7.

Q 2.5: For each of these instructions, finish the C macro expression:

- i) Test if the UDRE0 bit is 1: `if( UCSR0A & (1 << UDR20) ) {`
- ii) Write a 1 (set) to the TXEN0 bit: `UCSR0A |= (1 << TXEN0);`
- iii) Write a 0 (clear) to the UCPOL0 bit: `UCSR0A &= ~ (1 << UCPOL0);`

## Part 3: What Data to Send?

In the datasheet, we can find information about the UDR0 register, and how it acts as a buffer for transmitting and receiving data. In order to send data through UART, we just need to write some data to this register. Once you write data to UDR0, the USART peripheral immediately starts sending that data according to

the UART protocol, with the settings that you have already initialised the peripheral to.

Before we can do that, we probably need to store the data we are sending in a variable of some sort, and then set the UDR0 register to the value of that variable.

*Q 3.1: On the ATmega328P, how many bits of data can be stored in a variable of each of these types:*

char: 8 bits      uint8\_t: 8 bits  
 int: 16 bits      uint16\_t: 16 bits  
 int8\_t: 8 bits      float: 32 bits

*Q 3.2: What is the size of the UDR0 register?* 8 bits

*Q 3.3: What is the largest number we can transmit?* 255

Given these limitations, it might seem like we can't send very much at once. There's another element to consider though – we have already talked about how we need both the transmitter and receiver to agree on a set of parameters that dictate how the data is formatted. Actually, this only determines the physical layer – how are the bits physically sent and received. It does not consider what those bits actually mean! At this stage, they are just a bunch of 0s and 1s.

Of course, we could just interpret these as integers. But what if we wanted to send a word, or a song, or a picture? Then we need *character encoding*, which is like having a dictionary that says certain integers should mean different things. You can also think of this like a truth table – if the input is integer X, then the output should be data Y. One of the most popular encodings is ASCII:

| Decimal | Hex | Char                   | Decimal | Hex | Char    | Decimal | Hex | Char | Decimal | Hex | Char  |
|---------|-----|------------------------|---------|-----|---------|---------|-----|------|---------|-----|-------|
| 0       | 0   | [NULL]                 | 32      | 20  | [SPACE] | 64      | 40  | @    | 96      | 60  | `     |
| 1       | 1   | [START OF HEADING]     | 33      | 21  | !       | 65      | 41  | A    | 97      | 61  | a     |
| 2       | 2   | [START OF TEXT]        | 34      | 22  | "       | 66      | 42  | B    | 98      | 62  | b     |
| 3       | 3   | [END OF TEXT]          | 35      | 23  | #       | 67      | 43  | C    | 99      | 63  | c     |
| 4       | 4   | [END OF TRANSMISSION]  | 36      | 24  | \$      | 68      | 44  | D    | 100     | 64  | d     |
| 5       | 5   | [ENQUIRY]              | 37      | 25  | %       | 69      | 45  | E    | 101     | 65  | e     |
| 6       | 6   | [ACKNOWLEDGE]          | 38      | 26  | &       | 70      | 46  | F    | 102     | 66  | f     |
| 7       | 7   | [BELL]                 | 39      | 27  | '       | 71      | 47  | G    | 103     | 67  | g     |
| 8       | 8   | [BACKSPACE]            | 40      | 28  | (       | 72      | 48  | H    | 104     | 68  | h     |
| 9       | 9   | [HORIZONTAL TAB]       | 41      | 29  | )       | 73      | 49  | I    | 105     | 69  | i     |
| 10      | A   | [LINE FEED]            | 42      | 2A  | *       | 74      | 4A  | J    | 106     | 6A  | j     |
| 11      | B   | [VERTICAL TAB]         | 43      | 2B  | +       | 75      | 4B  | K    | 107     | 6B  | k     |
| 12      | C   | [FORM FEED]            | 44      | 2C  | ,       | 76      | 4C  | L    | 108     | 6C  | l     |
| 13      | D   | [CARRIAGE RETURN]      | 45      | 2D  | -       | 77      | 4D  | M    | 109     | 6D  | m     |
| 14      | E   | [SHIFT OUT]            | 46      | 2E  | .       | 78      | 4E  | N    | 110     | 6E  | n     |
| 15      | F   | [SHIFT IN]             | 47      | 2F  | /       | 79      | 4F  | O    | 111     | 6F  | o     |
| 16      | 10  | [DATA LINK ESCAPE]     | 48      | 30  | 0       | 80      | 50  | P    | 112     | 70  | p     |
| 17      | 11  | [DEVICE CONTROL 1]     | 49      | 31  | 1       | 81      | 51  | Q    | 113     | 71  | q     |
| 18      | 12  | [DEVICE CONTROL 2]     | 50      | 32  | 2       | 82      | 52  | R    | 114     | 72  | r     |
| 19      | 13  | [DEVICE CONTROL 3]     | 51      | 33  | 3       | 83      | 53  | S    | 115     | 73  | s     |
| 20      | 14  | [DEVICE CONTROL 4]     | 52      | 34  | 4       | 84      | 54  | T    | 116     | 74  | t     |
| 21      | 15  | [NEGATIVE ACKNOWLEDGE] | 53      | 35  | 5       | 85      | 55  | U    | 117     | 75  | u     |
| 22      | 16  | [SYNCHRONOUS IDLE]     | 54      | 36  | 6       | 86      | 56  | V    | 118     | 76  | v     |
| 23      | 17  | [ENG OF TRANS. BLOCK]  | 55      | 37  | 7       | 87      | 57  | W    | 119     | 77  | w     |
| 24      | 18  | [CANCEL]               | 56      | 38  | 8       | 88      | 58  | X    | 120     | 78  | x     |
| 25      | 19  | [END OF MEDIUM]        | 57      | 39  | 9       | 89      | 59  | Y    | 121     | 79  | y     |
| 26      | 1A  | [SUBSTITUTE]           | 58      | 3A  | :       | 90      | 5A  | Z    | 122     | 7A  | z     |
| 27      | 1B  | [ESCAPE]               | 59      | 3B  | ;       | 91      | 5B  | [    | 123     | 7B  | {     |
| 28      | 1C  | [FILE SEPARATOR]       | 60      | 3C  | <       | 92      | 5C  | \    | 124     | 7C  |       |
| 29      | 1D  | [GROUP SEPARATOR]      | 61      | 3D  | =       | 93      | 5D  | ]    | 125     | 7D  | }     |
| 30      | 1E  | [RECORD SEPARATOR]     | 62      | 3E  | >       | 94      | 5E  | ^    | 126     | 7E  | ~     |
| 31      | 1F  | [UNIT SEPARATOR]       | 63      | 3F  | ?       | 95      | 5F  | _    | 127     | 7F  | [DEL] |

In the table, the numbers on the left are the integer values that correspond to the character outputs on the right in each column.

Q 3.4: What decimal numbers correspond to the word "HELLO"? 72, 69, 71, 76, 79.

ASCII is a very old character encoding standard (from the 1960s!), originally developed for telegraphs. This is why there are a bunch of characters in the first column that don't seem to mean much, and are largely ignored in modern computing. You might notice that the decimal numbers go up to 127 – this is intentional, as it allows us to represent the characters within one byte.

Most *terminal* programs speak ASCII. When you send data from your ATmega328P /ATmega328PB to a computer or a phone, if using a terminal program on the receiving end it is going to try to decode the numbers that you send with ASCII. This means that if you want the terminal program to show the number 63, then you can't just send the number 63 – you have to send a 54 (i.e. "6") followed by a 51 (i.e. "3") in order to display the characters 6 and 3.

Q 3.5: Fill in the blanks below to describe how to send a number to a terminal:

We can extract the individual digits of a number by using the modulo operator. Then we encode an individual digit into its ASCII equivalent by simply adding the decimal number 48. Then we iterate through each digit of the number using a loop, writing each character to the UDR0 register.

Note: While terminal programs generally use ASCII, in some projects, you may find that you want to use other encodings – even something custom!

## Part 4: Transmitting Data

Now that we understand both how to control the USART peripheral and how to format our data correctly, we can send some data to another computer. We will need to write some C code that does this. Our eventual goal is to send the list of prime numbers from the pre-lab to the terminal on a computer, but let's break that up into a few smaller tasks.

Q 4.1: Assuming that we can begin transmitting a UART packet immediately after the previous one ends, work out the length of time required to send each of these:

A single character: \_\_\_\_\_

A 3-digit number: \_\_\_\_\_

Q 4.2: Now, also assuming that we need to send a comma character and a space character between each number, work out the length of time to send each of these:

Three 3-digit numbers (there are 2 commas and 2 spaces): \_\_\_\_\_

The entire primes list from pre-lab (assume all 3-digits): \_\_\_\_\_

Our C code will have two main parts: initialisation and transmission. The initialisation is only run once, and configures the USART peripheral to behave the way we need it to. The transmission part is run as many times as needed. Therefore, it probably makes sense to turn this into two functions. Modify the Atmel Studio project you created to answer the pre-lab questions to answer the following questions.

*Q 4.3: Write a `usart_init(uint16_t ubrr)` function which sets up the USART peripheral as determined in Part 2. Of the five control registers, how many could be left with their initial values? \_\_\_\_\_*

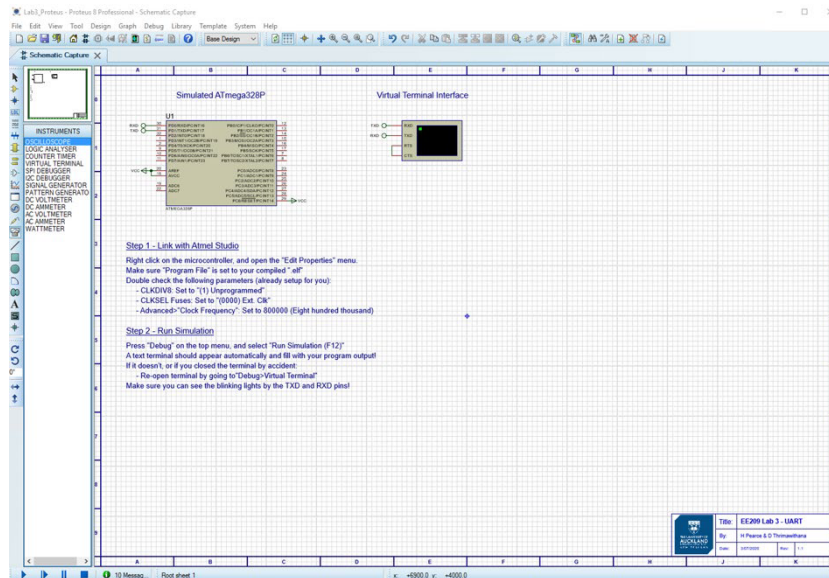
*Q 4.4: Write a `usart_transmit(uint8_t data)` function which handles transmission of a single number through the USART peripheral. The important steps are to:*

- 1) Check bit \_\_\_\_\_ of register \_\_\_\_\_ and wait if \_\_\_\_\_*
- 2) Load \_\_\_\_\_ variable into the \_\_\_\_\_ register*

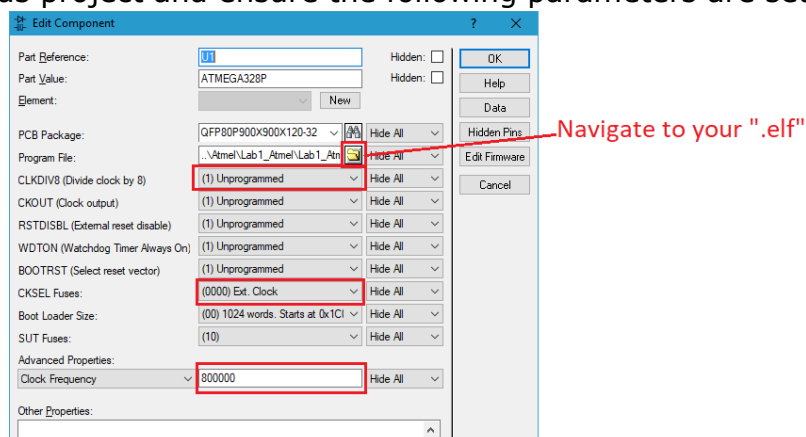
*Q 4.5: Construct a program that will allow you to send the character "3" to be printed on a terminal program window every 0.5s infinitely, following these steps:*

- 1) Call `usart_init(ubrr)` using the UBRR value calculated in Q2.2*
- 2) In the `while(1)` loop, every 0.5s call the `usart_transmit(character)` function, where character is the ASCII value for the character "3". You can use `<util/delay.h>` and `_delay_ms(time)` to create the 0.5s delay.*

Once your code is complete, we can test it in Proteus. First let's turn-on optimisation for the Atmel Studio project. Go to Project > [ProjectName] Properties > Toolchain > AVR/GNU C Compiler > Optimization, and set the Optimization level to "Optimize -O1". Then make sure to compile your code in Atmel Studio. Now navigate to your Atmel Studio project folder using Windows Explorer and find the folder called "debug". Inside there will be two compiled files, a ".hex" and a ".elf". These represent the compiled binary of your project code – your actual program, in machine code! When you download your program to the real board, these are the files that would be sent. Proteus can use these files in simulation. In your local lab repository, go to the folder "Lab2\_UART\Lab2\_Proteus", then open the "Lab2\_Proteus.pdsprj" file. You should see something like the following:



This Proteus project is made up of only two components, the emulated ATmega328P microcontroller, and the virtual terminal interface. To link the Atmel Project and the Proteus projects, double click on the ATmega328P microcontroller in the Proteus project and ensure the following parameters are set:



Program File: Set to your compiled ".elf" in the "debug" directory (from earlier)  
– oppose to ".hex" the ".elf" file contains debug information.

CLKDIV8: Set to "(1) Unprogrammed"

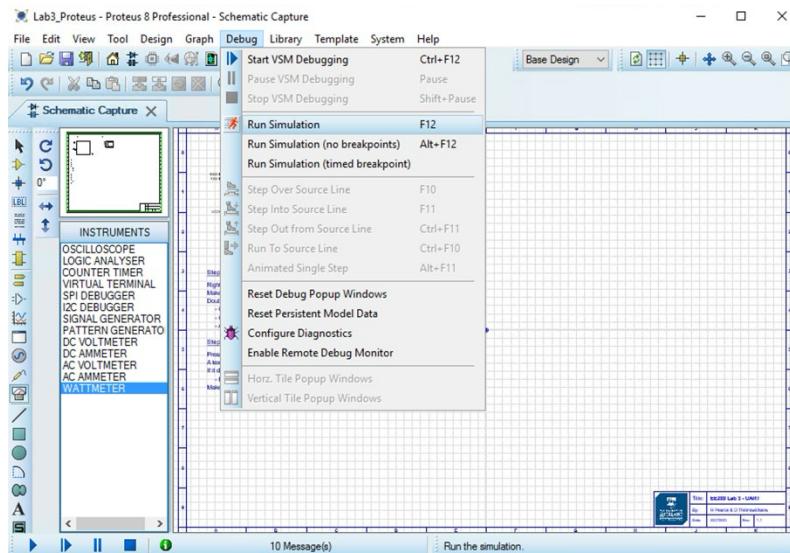
CLKSEL Fuses: Set to "(0000) Ext. Clock"

Advanced Properties>Clock Frequency: Set to 2000000 (i.e. 2MHz)

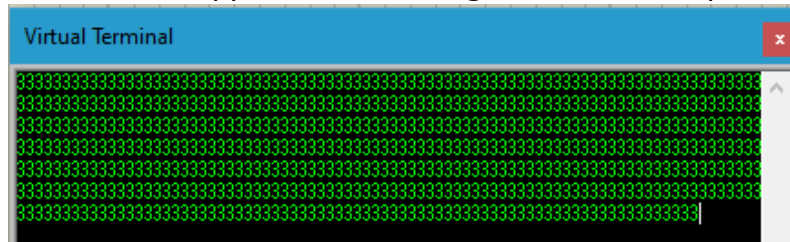
What we have now done is set the virtual microcontroller to run at 2MHz. We choose this speed because the Xplained Mini 328P/328PB boards you will use during this course and in the project is setup to use a 2MHz system clock (as per design specifications).

Note that the Xplained Mini boards can be configured to run at 16MHz (this is in fact the default setting). However, the Xplained Mini boards given to you have been modified according to your design specifications to run at 2MHz (by setting the CKDIV8 fuse bit).

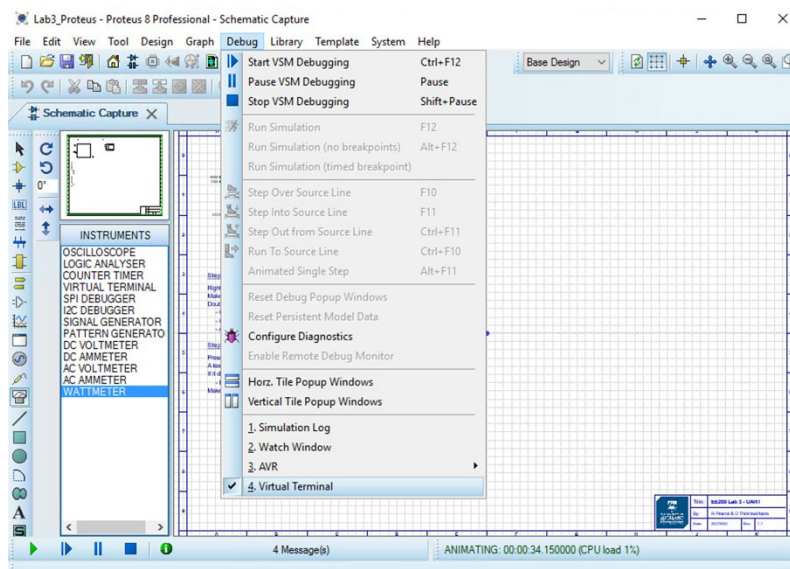
Now let's run the simulation. On the top menu, press Debug>Run Simulation (alternatively you can press F12 or the blue play button found in lower left corner):



A terminal window should appear, and the digit "3" should repeatedly fill it:



If your terminal window didn't appear after you start the simulation, or you close it accidentally, then, while a simulation is running, you may recall it by pressing Debug>Virtual Terminal:



If your terminal is present but empty, check that you've correctly initialised the UART peripheral and setup the transmit function in your Atmel Studio project. If your terminal is showing garbled characters check; baud rate register values are correct, control registers are correctly setup and system clock in Proteus is set to 2000000.

Press Debug>Stop Simulation when you're finished (alternatively you may press the blue stop button found in lower left corner).

**BACK TO WORK**

Let's do something a bit more complicated now – transmitting a number with multiple digits. We need to separate these out into individual characters, as described in Part 3 of this lab.

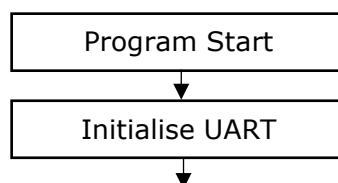
*Q 4.6: Modify your program so that you transmit the number "345" to the virtual terminal on Proteus every 0.5s. Inside the while(1) loop, implement your answer from Q3.5 in code, so that the individual 3, 4, and 5 characters are extracted and transmitted to display 345 every 0.5s. If you wish, you may add a space too.*

Once you've got this working, it's time to progress to the main task – transmitting a long list of prime numbers! For this lab, we want to transmit each of those numbers so that they appear on a terminal screen, with a comma and a space after each number. To make this task easier, you may print all prime number as 3-digit numbers (e.g., prime number '2', can be printed on terminal as '002')

*Q 4.7: Modify your program so that it follows these steps:*

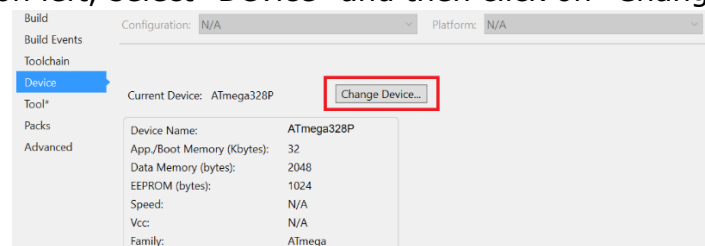
- 1) Call `usart_init(ubrr)` using the UBRR value calculated in Q2.2
- 2) Using the pre-lab code, create an array with the prime numbers
- 3) Inside the while(1) loop, set up a \_\_\_\_\_
  - a. Extract the individual characters of each prime number
  - b. Call `usart_transmit(character)` as needed
  - c. Call `usart_transmit(character)` for the comma and space as needed
  - d. Increment \_\_\_\_\_

*Q 4.8: Complete the flowchart below that describes the flow of your program code:*



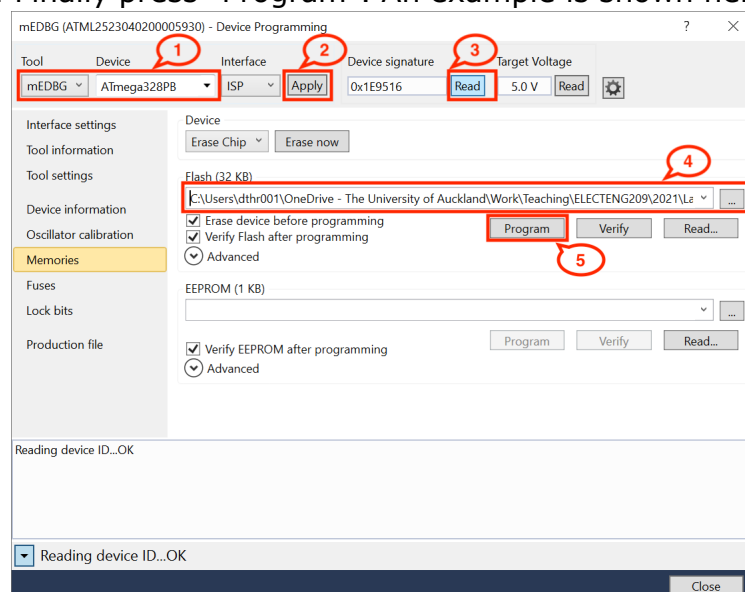
## Part 5: Programming the Xplained Mini

Now that we have developed our program and verified its functionality using Proteus, we are ready to load it on to the Xplained Mini 328P/328PB. Depending on availability, most of you will be provided with a Xplained Mini 328P this year. However, as the semester progress and when we replace damaged boards we may have to issue you a Xplained Mini 328PB. In case you have a Xplained Mini 328PB, remember to recompile the code to run on an ATmega 328PB (so far we have been compiling the code to run on an ATmega328P). To do this, in Atmel/Microchip Studio, go to Project>'Project Name' Properties (here 'Project Name' will be the name you gave your Atmel/Microchip Studio project). A new tab will open and from the menu on left, select "Device" and then click on "Change Device":



A window will open allowing you to pick a microcontroller. Select ATmega328PB as the microcontroller and click "Ok". Now your project is modified so that it will compile the code to run on the Xplained Mini, which has a ATmega328PB on board. Save the project and "Build" your project again.

Connect the Xplained Mini 328P/328PB to your PC using a USB cable. When doing for the first time it may take a few minutes to install drivers. Make sure to wait until drivers are installed to avoid damaging the board. Now to load the recompiled ".elf" file to the ATmega328P/ATmega328PB on the Xplained Mini, bring up the device programmer from Tools>Device Programming. Select "Tool" as "mEDBG ..." and "Device" as "ATmega328P" or "ATmega328PB" depending on the device you have. Press "Apply". To check if your Xplained Mini can now communicate with Atmel/Microchip Studio, click "Read" under "Device Signature". If everything is well it will show device signature as a hex number. Now click on "Memories" and check if the file name under "Flash" is set to the ".elf" file (normally this is setup automatically). Finally press "Program". An example is shown here:



Now on the PC open a terminal program like PuTTY or RealTerm. Setup this terminal program to received the messages sent by the Xplained Mini by telling it the baud rate (i.e., 9600), the number of data bits (i.e., 8), the number of stop bits (i.e., 1) and the serial port number. You can find the serial port number using the "Device Manager". You should be able to now see the prime numbers printed on the terminal.

You should not put the Xplained Mini in to the debugWIRE mode. If you do then stop debug and disable the debugWIRE mode before closing to avoid damaging the Xplained Mini.

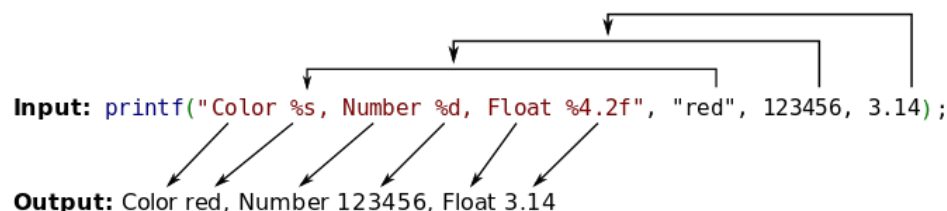
## Optional Task: Advanced String Processing

In this lab you broke apart numbers into their individual digits and converted them manually into ASCII. However, this is not the only way to do this process! You may have learned in ENGGEN101 about the *printf()* function in C, stored in `<stdio.h>`, which lets you format strings to the standard output in C, e.g.:

```
#include <stdio.h>

int main() {
    printf("Hello, world!\r\n");
}
```

We can use *printf*, and the closely related *sprintf()* on our microcontroller as well! Let's first have a refresher on how these functions work. Firstly, both *printf* and *sprintf* take a *format string* which defines the output, and then a varying number of arguments to be passed in as the values to be placed in the format string. Format specifiers stating with % sign tell where to put these arguments and how to format them. For example:



There are many format specifiers, but the most important are the following:

| Specifier | Used for              | Example                                                                                            |
|-----------|-----------------------|----------------------------------------------------------------------------------------------------|
| %c        | Individual characters | <pre>char c = 'a'; printf("Char 1: %c", c);</pre> <p>→ Char 1: a</p>                               |
| %s        | Strings of characters | <pre>char *name = "Hammond"; printf("My name is %s", name);</pre> <p>→ My name is Hammond</p>      |
| %d        | Integers              | <pre>int x = 10; printf("The number is %d", x);</pre> <p>→ The number is 10</p>                    |
| %f        | Floats                | <pre>Float e = 2.71828 Printf("Euler's number is %f", e);</pre> <p>→ Euler's number is 2.71828</p> |

You may also place arguments around the format specifiers for the purposes of rounding as well as leading spaces and zeros on numbers (if you want to produce a neatly formatted table, for instance). The full format specifier prototype is `%[flags][width][.precision][length]specifier`; and you can find more detail on this at [link](#). Note that Atmel/Microchip Studio is set up to using a lightweight string printing library by default which doesn't support float formatting.

The function `printf` writes the output of the string to your *standard output*. On a computer, this is the terminal. On a microcontroller, we would need to define exactly what our "standard output" is. However, it's easier, and tidier, to use `sprintf` instead. When using `sprintf` the output is written into a character array, which we can then use however we'd like. For instance:

```
//we need to #include <stdio.h>
//and
#include <string.h>

char* name = "Hammond";
char string_buffer[30]; //we can create strings up to 30 characters long

//this function creates "My name is Hammond" and stores it into string_buffer
sprintf(string_buffer, "My name is %s\n", name);

//we can now send string_buffer to the Uart
for(int i = 0; i < strlen(string_buffer); i++)
{
    USART_Transmit(string_buffer[i]);
}
```

This code will create a nicely formatted string, and save it into the character array called "string\_buffer". We can then iterate over that array, sending the characters one by one to our USART peripheral. Note if using above code we also need to include `<string.h>` for the `strlen` function as indicated in the comments.

*Q O.1) Save your code from earlier to a different project. Now, rewrite your code from Q 4.6 to format the prime numbers as strings using `sprintf` before sending them to the UART. Note that you now no longer need to perform the manual conversion to ASCII – `sprintf` handles this for you automatically!*

`Printf` and `sprintf` have a major downside when operating on a simple device like a microcontroller. Consider the complexity that `printf/sprintf` have inside them in order to achieve all those formatting functions!

*Q O.2) Compile both the `sprintf` and the original version of your code. Now, compare the size of the generated code. Which is larger? \_\_\_\_\_ By how many times is it larger? \_\_\_\_\_*

**To get signed off for the lab:**

- Record all the workings in hardcopy or in digital format so we can give you marks for workings if final answers are not correct
- Modify the simulation models provided as per the lab tasks and save
- Commit and push your saved work to the lab repository on GitHub after completing each task and make sure it is up to date
- Complete this document summarising your final answers, commit and push to lab repository on GitHub
- Update logbook indicating yours and teams progress, meeting notes, etc.
- Go to your assigned interview session to check-in with a TA who will check your solutions and ask a few questions
- Report on your weekly progress